

# AWS AI Job Search Agent

Profile-Isolated Job Discovery, Governance, ML Ranking & GenAI Review

AWS Portfolio Project

Employer Adapters | Deterministic Governance | SageMaker | Amazon Bedrock | S3 Website | LangSmith

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Jingru Chen • September 29, 2026

One shared AWS platform while search, ranking, publication and history remain isolated by candidate profile.

## BUSINESS PROBLEM

- Employer sites expose different search backends and schemas.
- Candidate profiles require different job families and screening rules.
- Daily runs must avoid rescoring recent postings while still scoring every NEW eligible job.
- Recruiter-facing output needs traceable AWS / AI evidence.

## ARCHITECTURE RESPONSE

- Shared platform + employer adapter layer.
- Central profile configuration; no title logic duplicated inside employer code.
- Deterministic screening before ML / GenAI.
- Profile-aware lifecycle and score cache.
- Daily publication with 7-day history.

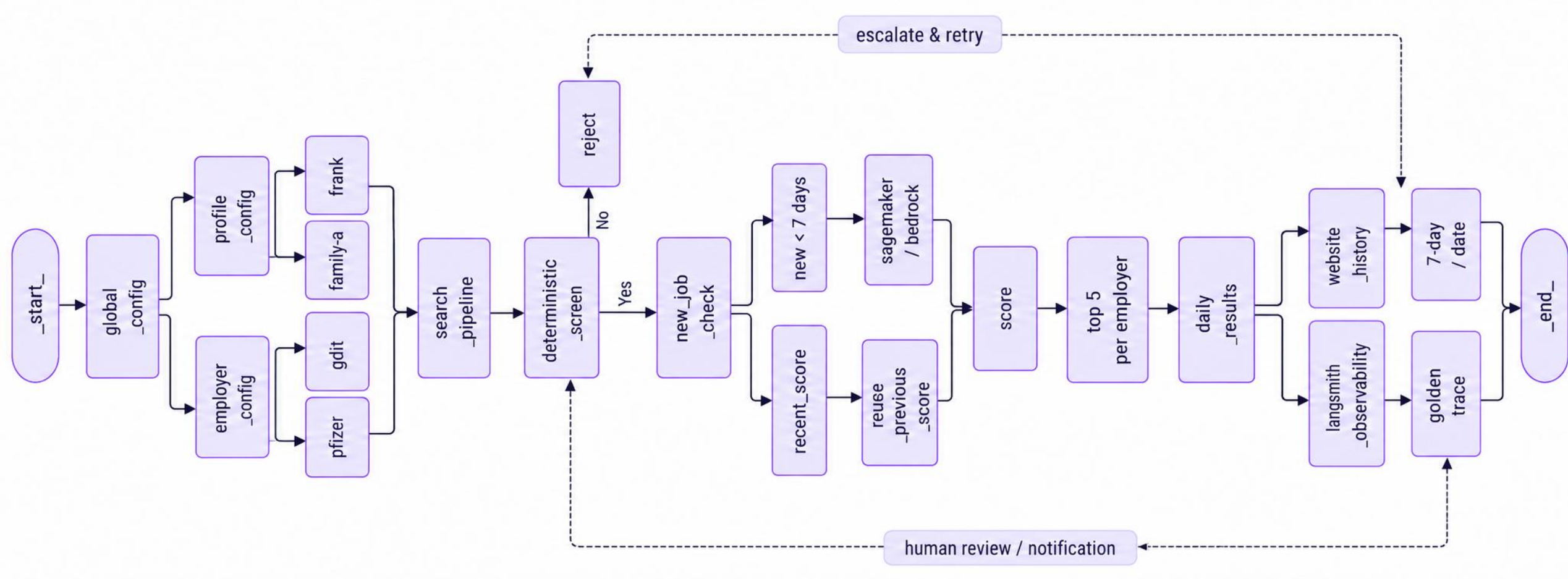
## PROFILE ISOLATION

Frank C.  
Data Science • AI • Quant • Analytics

W.Y. / Family-A  
Bioanalysis • LC-MS/MS • Mass Spectrometry

Rule: profiles share infrastructure, not business search results.

Design principle: share infrastructure → isolate business intent → govern deterministically → score incrementally → publish auditable daily results.



Incremental processing reduces unnecessary inference while preserving profile isolation and deterministic controls.

## 1 DISCOVER

Employer adapters collect postings; profile configuration defines allowed job families.

## 2 GOVERN

Exclude internships, irrelevant roles and disallowed clearance requirements before AI scoring.

## 3 RESOLVE

Identify NEW eligible postings; reuse recent cached scores when allowed by lifecycle policy.

## 4 SCORE

Score every NEW eligible posting with SageMaker; retain model evidence for downstream selection.

## 5 REVIEW

Use Bedrock for grounded review while preserving deterministic governance boundaries.

## 6 PUBLISH

Rank high-to-low and retain up to Top 5 NEW jobs per employer; publish profile/date artifacts.

Daily contract: score all NEW eligible jobs first → then select Top 5 per employer → publish without cross-profile leakage.

Two profile-specific SageMaker XGBoost models provide quantitative job-fit scoring; Amazon Nova 2 Lite on Bedrock provides grounded semantic review.

**1 Human / graded  
Ground Truth**

**2 Structured  
Features**

**3 XGBoost  
Training**

**4 SageMaker  
Inference**

**5 Job-Fit  
Score**

**6 Nova 2 Lite  
GenAI Review**

**7 Governed  
Ranking**

## FRANK C. — XGBOOST JOB-FIT MODEL

- Supervised XGBoost model trained in SageMaker.
- Ground truth: 24 labeled training rows; 8 human-approved positive rows.
- Structured fit features feed quantitative ranking.
- Real Batch Transform example: GDIT RQ227420 → score 0.8923.
- Interpretation guardrail: small-sample demonstration score, not a calibrated probability.

## FAMILY-A / W.Y. — XGBOOST MODEL

- Separate profile-specific SageMaker XGBoost model for bioanalysis roles.
- Training evidence includes a graded multiclass training dataset.
- Independent model + inference + prediction artifacts preserve profile isolation.
- Purpose: rank LC-MS/MS / bioanalysis opportunities using Family-A-specific relevance.
- Exact aggregate accuracy / calibration metrics are not claimed by the recovered evidence.

## AMAZON BEDROCK — NOVA 2 LITE

- Amazon Nova 2 Lite is a transformer-based generative foundation model available through Amazon Bedrock.
- Project role: read JD + profile context and produce grounded semantic review.
- Outputs: verified matches, gaps / unknowns, concise assessment.
- Bedrock does not replace deterministic screening and does not overwrite the SageMaker score.
- Separation of duties: XGBoost ranks quantitatively; Nova explains semantically.

ML / GenAI design: human-review labels -> profile-specific XGBoost scoring -> grounded Nova 2 Lite review -> governed publication

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

jxc518 (288229085061)

Amazon SageMaker AI

Training jobs

job-search-frank-fit-20260928012513

Amazon SageMaker AI

Dashboard

What's new 41

Environment configuration

Applications and IDEs

Model training & customization

job-search-frank-fit-20260928012513

Clone

Create model package

Stop

Create model

Job settings

Job name

Status

SageMaker metrics time series

IAM role ARN

ARN

Creation time

Last modified time

Training time (seconds)

Billable time (seconds)

Managed spot training savings

Tuning job source/parent

Algorithm

Algorithm ARN

Additional volume size (GB)

Maximum wait time for managed spot trainings

Volume encryption key

Training image

Maximum runtime (seconds)

Managed spot training

Input mode

Instance group

Instance type

Instance count

Keep alive period

[Alt+S]

United States (N. Virginia)

jxc518 (288229085061)

Models

Create endpoint

Create endpoint configuration

Actions

Create model

Find models

1

Models table

Bedrock Marketplace (152)

Looking for latest open-weight and frontier models from Anthropic, OpenAI, and other providers? Click here to go to the new console optimized for bedrock-mantle and choose the region where the model is available.

Try the new Bedrock Console

Learn about bedrock-mantle endpoint

Spotlight

Find models

1 match

Recently released

Model name = Nova 2 Lite

Clear filters

Nova 2 Lite

By Amazon

Image, Video, Text to Text

Serverless

Cross-region inference

HR-ready website presents candidate-specific daily history while keeping the underlying platform shared.

## FRANK C. — DATA SCIENCE / AI

September 29 daily publication

- Root summary: 3,154 raw jobs searched
- 191 passed deterministic governance
- 191 AI-scored NEW jobs
- 20 published Top jobs

Employers represented in the frozen daily artifact include Capital One, GDIT, Navy Federal and Optum.

## W.Y. / FAMILY-A — BIOANALYSIS

September 29 daily publication

- Real employer/job records only
- AstraZeneca — Oncology Bioanalytical Strategy Lead
- Pfizer — Senior Scientist Mass Spectrometrists

September 28 was intentionally not backfilled because the only recovered Family-A artifact was a synthetic contract-test fixture.

Governance choice: missing historical evidence is preferable to publishing synthetic or unverifiable job data.

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Application

All applications

Search

Home

Tracing34

EnginePlus

Monitoring1

Datasets & Experiments3

Evaluators

Annotation Queues2

Custom Apps

Prompts

Playground

Studio

Context Hub

DeploymentsPlus

Sandboxes

LLM GatewayPlus

Keyboard Shortcuts

Notifications1

Settings

Personal

chen.jingru@gmail.com

Personal / Tracing / AWS-AI-Job\_Search\_Agent / V2 Integration Validation - AWS-...

V2 Integration Validation - AWS-AI-Job\_Search\_Agent

Trace

Waterfall

Filter runs...

0s

1s

2s

3s

4s

5s

V2 Integration Validation - AW... 5.54s

01 Profile Configuration 5.08s

Load Profile Configuration 4.95s

02 Search Planning 4.82s

Build Search Plan 4.68s

Resolve Employer Set 4.54s

03 Company Collection 4.41s

Employer 01 | astrazeneca 4.27s

Employer 02 | gdit 4.13s

Employer 03 | pfizer 4.00s

Employer 04 | gdit 3.86s

Employer 05 | optum 3.73s

Employer 06 | pfizer 3.57s

04 Canonical Normalization 3.43s

Schema Validation 3.31s

Job Identity Resolution 3.18s

05 Governance 3.03s

Clearance Classification 2.90s

Job Family Policy 2.75s

V2 Integration Validation - AWS-AI-Job\_Search\_Agent

Feedback

Input

Output

Attributes

Feedback

+ Add feedback

Input

Fields

artifact\_type DAILY\_PLATFORM\_COMPLETION

completion\_id 0296af3d377274268a24835887ecad5e9a2ac19b2f5ec5e7b83b516d9...

observer\_mode READ\_ONLY\_ARTIFACT\_OBSERVER

snapshot\_date 2026-09-29

Output

Fields

stage V2 Integration Validation - AWS-AI-Job\_Search\_Agent

status OBSERVED\_FROM\_ARTIFACT

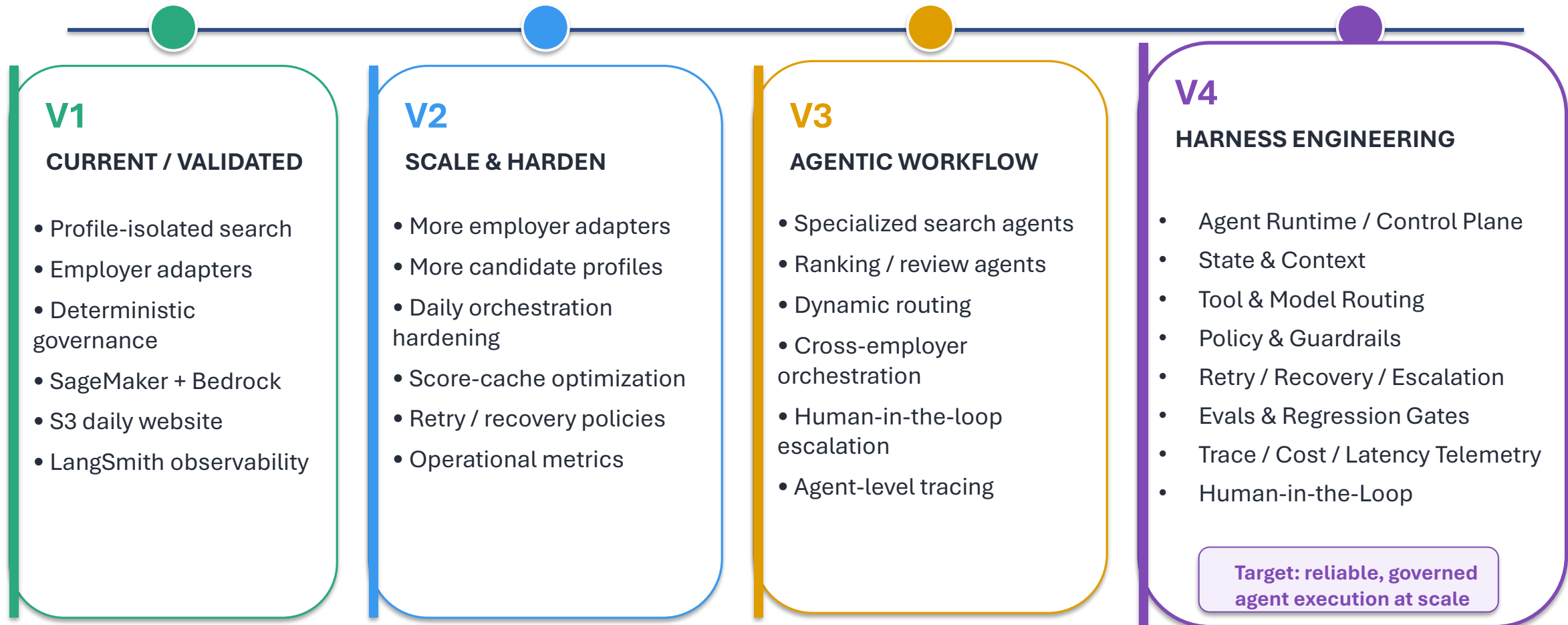
Attributes

Metadata

LangSmith

ls\_run\_depth0

Additive evolution from the frozen V1 baseline—scale first, then agentic orchestration, then production-grade harness engineering.



## Key Engineering Decisions

Adapter-based employer integration • Profile-isolated configuration • Deterministic-before-AI governance • Incremental score reuse • Separate ML ranking / GenAI review • Artifact-driven observability